

Project 6: The Sharpness Quest. Does a Discriminator Loss Sharpen a cVAE?

Group 4

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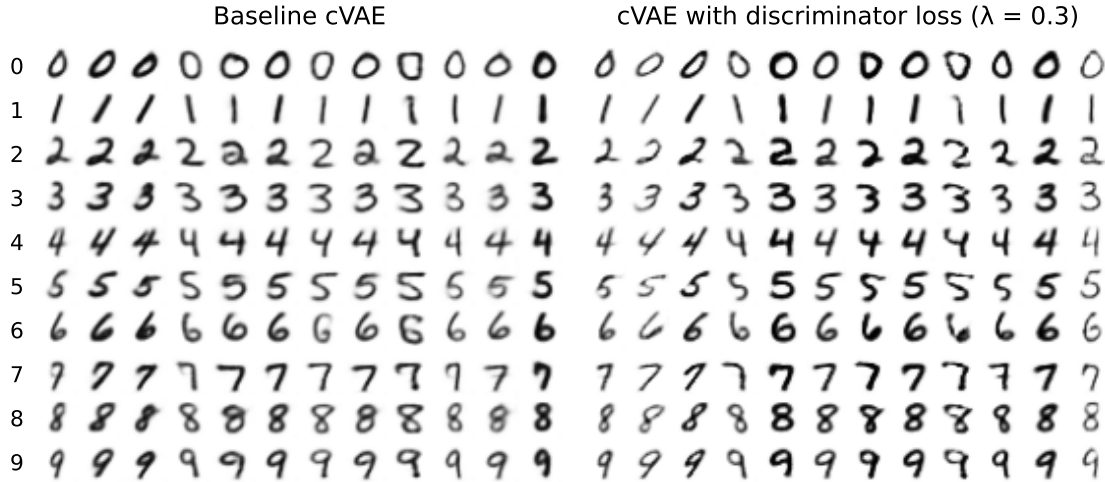


Figure 1: Twelve samples per digit from the baseline cVAE (left) and the cVAE with a discriminator loss (right). Both decode the *same* twelve latent codes, one per column. The classifier’s accuracy on each grid is 1.0000 (left) and 0.9917 (right).

1. Introduction

A conditional variational autoencoder (cVAE; Kingma and Welling, 2014; Sohn et al., 2015) encodes an image with its class into a latent code z and decodes it back; decoding a draw from $\mathcal{N}(0, I)$ under a chosen class generates a new digit. Trained on pixel-wise mean squared error (MSE), it blurs: when several images fit one code, the MSE-optimal output is their mean, a grey smear of differently placed strokes. A discriminator (Goodfellow et al., 2014), a classifier trained to tell real images from generated ones, penalises any output that does not look like data, such as that smear. Hence the hypothesis tested here: “adding a discriminator term to the loss function of a cVAE during training increases the sharpness of the reconstructed images”.

Prior work supports it, mostly by eye. Larsen et al. (2016) trained a VAE jointly with a GAN (VAE/GAN) and found its samples, and its reconstructions of held-out CelebA faces, sharper than a plain VAE’s. The conditional CVAE-GAN of Bao et al. (2017) produced images that “look realistic and clear” where a CVAE’s were blurry. Rosca et al. (2017) added adversarial terms to an autoencoder (α -GAN) and obtained better reconstructions than their autoencoder baseline, but mixed sample quality. Dosovitskiy and Brox (2016) found that image-space losses over-smooth where adversarial ones stay crisp. Those studies change several things at once; here the discriminator term is the only change, and sharpness is measured.

2. Methodology

Baseline cVAE. Tutorial 8.3’s model has a convolutional encoder and a mirrored decoder, both conditioned on a one-hot class, and the brief’s 7 latent dimensions; it trains on $100 \text{ MSE} + 0.1 \text{ KL}$ with AdamW (learning rate 0.001) in batches of 1,000. For the 20×20 images the encoder’s last kernel and the decoder’s first become 3×3 , the only architectural change (350,107 parameters). At the tutorial’s constant rate the loss kept falling long after its 50 epochs, so both models follow a cosine schedule from 0.001 to 0 over 150 epochs.

Discriminator loss. A conditional discriminator D

(two strided convolutions and a linear layer, 23,297 parameters) sees the image with its class as ten constant channels. As Salvado (2026, p. 41) prescribes, real images are labelled 1 and fakes 0, D minimises binary cross-entropy, and each batch takes a D step, then a cVAE step. The fakes are reconstructions $\text{Dec}(z, c)$, which the hypothesis names, and prior samples $\text{Dec}(z_p, c)$, which Task 3 evaluates, at half weight each, as in Larsen et al. The cVAE minimises $100 \text{ MSE} + 0.1 \text{ KL} + \lambda L_G$ with $L_G = -\log D(\text{fake})$, the non-saturating form of the brief’s $\log(1 - D(G(z)))$: it shares the fixed point but keeps its gradient while D confidently rejects early fakes (Goodfellow et al., 2014, §3). L_G sees z detached, so only the decoder learns to fool D , again as in Larsen et al. D uses AdamW (learning rate 2×10^{-4}) on the same schedule.

Protocol. A seeded random 10,000 of the 60,000 training images form a development split. There, λ was chosen from $\{0.03, 0.1, 0.3, 1.0\}$ as the value whose reconstructions came closest to the real development images’ Laplacian variance (0.2311), excluding runs where D passed 98% accuracy. In that order, the four runs’ reconstructions scored 0.1466, 0.1732, 0.1909 and 0.1857, with D accuracies of 0.8749, 0.7299, 0.6331 and 0.5997, so $\lambda = 0.3$ was chosen. Training counts as converged when the mean development reconstruction loss over the last 10 epochs is within 1% of the previous 10; both runs meet it (changes of -0.0013 and -0.0012 ; Figure 2b). The final models were retrained on all 60,000 images with seeds 0, 1 and 2, and only `main_report.ipynb` scores the test split.

3. Experiments

Each of the 10,000 test images is reconstructed from its encoder mean, and one prior sample per image is drawn under its label from the same code for both models, so all comparisons are paired. **Sharpness** is the variance of the Laplacian (Pech-Pacheco et al., 2000), read against the real images’ own 0.2277, since noise raises it too. The **mid-grey fraction** (pixels strictly between 0.1 and 0.9) is the check: a share of 0.700 of real pixels is exactly white, so blur shows as grey, and noise would raise both measures together. **Class consistency** uses a separate

Measure (10,000 test images)	Real	Baseline	Disc. loss	Δ	95% CI	Δ at seeds 1, 2
Reconstruction MSE	n/a	0.01109	0.01194	+0.00085	[+0.00080, +0.00090]	+0.00238, +0.00067
Laplacian variance, reconstructions	0.2277	0.1184	0.1931	+0.0747	[+0.0741, +0.0753]	+0.0728, +0.0810
Mid-grey fraction, reconstructions	0.1439	0.2084	0.1533	-0.0551	[-0.0556, -0.0547]	-0.0555, -0.0590
Laplacian variance, samples	0.2277	0.1141	0.2055	+0.0913	[+0.0904, +0.0923]	+0.0906, +0.0943
Mid-grey fraction, samples	0.1439	0.2113	0.1469	-0.0644	[-0.0651, -0.0637]	-0.0662, -0.0672
Classifier accuracy, samples	0.9836	0.9855	0.9875	seeds: 0.9855 0.9861 0.9879	0.9875 0.9877 0.9911	
Fréchet distance, samples	0.45	51.32	4.81	seeds: 51.32 56.71 52.93	4.81 7.14 4.43	
Fréchet distance, reconstructions	0.45	38.62	5.71	seeds: 38.62 40.55 39.66	5.71 7.18 4.89	
Within-class spread, samples	1	0.640	0.883	seeds: 0.640 0.639 0.633	0.883 0.865 0.879	

Table 1: Seed-0 models; Δ is discriminator loss minus baseline, with a paired bootstrap interval. In the lower block “Real” is the real digits’ accuracy, the real-against-real floor and 1 by definition, and seeds read baseline / discriminator loss.

classifier trained on the training images (0.9836 test accuracy), as Salvado (2026, pp. 42, 87) proposes. **Quality and variety** use the Fréchet distance (Heusel et al., 2017) in that classifier’s 128-dimensional features, since Inception-v3 was built for 299×299 colour photographs. It compares class-stratified test halves (4,999 and 5,001 images), so the real-against-real distance is a fair floor, and within-class feature spread, generated over real, tests for mode collapse. Differences are discriminator loss minus baseline, with 95% paired bootstrap intervals (2,000 resamples) at seed 0, repeated at seeds 1 and 2. **The rule was fixed in advance:** an object is sharper if its Laplacian variance rises with an interval above zero and at every seed, and its mid-grey fraction falls.

4. Discussion

In Figure 1 every baseline digit is recognisable but soft: its strokes are mostly grey and fray into halos, most visibly on the loops of 5, 6 and 8. With the discriminator loss the same codes decode to crisp, clean-edged strokes on a white background, so sharpness and overall quality improve together. Class consistency holds in both grids, and each column keeps its slant and style across all ten rows. Variety is kept too, and stroke weight ranges more widely, from hairline to bold, as in real handwriting; the within-class spread in Table 1 agrees.

For reconstructions the rule is met: Laplacian variance rises from 0.1184 to 0.1931 (Table 1; Figure 2a), with an interval above zero and differences of +0.0728 and +0.0810 at seeds 1 and 2, and the mid-grey fraction falls from 0.2084 to 0.1533. Both values remain below the real test images’ 0.2277, so the change moves reconstructions toward real sharpness, not past it. The samples meet it too: Laplacian variance rises from 0.1141 to 0.2055 (+0.0906 and +0.0943 at seeds 1 and 2), and the mid-grey fraction falls from 0.2113 to 0.1469. The price is pixel fidelity: reconstruction MSE rises from 0.01109 to 0.01194, beyond evaluation noise. Pixel error punishes small shifts (Larsen et al.): a crisp stroke slightly misplaced costs more than a blur over both places.

Both models’ samples are read as their intended class about as often as real digits are (0.9855 and 0.9875 against 0.9836). In the classifier’s feature space the samples’ Fréchet distance to real digits falls from 51.32 to 4.81, at every seed (the real-against-real floor is 0.45), and the reconstructions’ from 38.62 to 5.71. Within-class spread is 0.640 for the baseline and 0.883 with the discriminator loss (1 matches real variety), so the discriminator’s samples are more varied, not less.

Strengths and limitations. The comparison is controlled: the models share data, architecture, optimiser, schedule, seeds and batch order, and each claim rests on paired intervals, three seeds and a rule stated in advance. But three seeds bound training variance loosely (the MSE cost ranges from +0.00067 to +0.00238 across seeds), four values of λ sample the trade-off coarsely, Laplacian variance also rewards noise, one pixel of blur is a large part

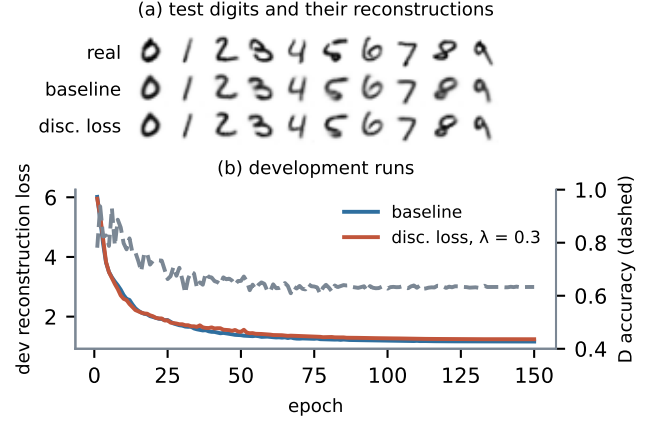


Figure 2: (a) One test digit per class, rebuilt by each seed-0 model. (b) Development reconstruction loss, with the discriminator’s accuracy dashed.

of a stroke at 20×20 , and the Fréchet distance uses one small classifier’s features. The discriminator also costs time: an epoch took 5.40 s against 2.09 s.

Recommendations. Use the discriminator term where perceived sharpness matters more than pixel accuracy, choosing λ on held-out data against the real images’ own sharpness rather than maximising it, which rewards noise; keep the adversarial gradient out of the encoder and anneal both optimisers, as here. Before generalising, add seeds, a finer λ grid, a perceptual measure from a network the models never saw, and natural images.

5. Conclusion

The evidence supports the hypothesis: under a rule fixed before the results, adding a discriminator term made both the reconstructions and the samples of a cVAE measurably sharper, consistently across three training seeds. Pixel error rose slightly, class consistency was unchanged and variety grew. Next, Larsen et al.’s shift-tolerant, feature-wise reconstruction loss would test whether the remaining blur comes from pixel MSE itself.

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